

Evaluating the Quality of Explanations under a Cognitive Feature Budget

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ABSTRACT

Local explanation methods are often evaluated with complete explanations, while users usually inspect only a few features. This project introduces Complexity-Calibrated Local Concordance (CCC), a functionally-grounded metric that tests whether an additive explanation remains faithful after top- K truncation. We implement an evaluation framework for tabular classification using two UCI datasets, two black-box models, and three explainers: LIME, SHAP, and MAPLE. The experiments show that SHAP is the strongest method under compact faithfulness, but also that exact full fidelity does not guarantee a faithful compact explanation. CCC therefore complements existing XAI metrics by making the trade-off between fidelity and human-sized explanations explicit.

KEYWORDS

Explainable AI, local explanations, faithfulness, cognitive complexity

1 INTRODUCTION

Explainable Artificial Intelligence (XAI) is used to make black-box predictions inspectable by users, developers, and domain experts. In tabular classification, local explanation methods such as LIME [10], SHAP [6], and MAPLE [9] are particularly common because they express each prediction as a set of feature contributions. The central evaluation problem is whether those contributions are faithful to the model rather than merely plausible to a human observer [4].

The EQE project (Evaluating the Quality of Explanations) focuses on a specific blind spot in this evaluation problem. Many faithfulness metrics evaluate the complete explanation, but complete explanations can contain all available features. A user-facing explanation, instead, should remain useful after being reduced to a small number of high-impact features. This is not only a design preference: Miller’s classic work on working-memory limits suggests that humans can actively process only a small number of information units at once [7]. For this reason, the project evaluates explanations under a feature budget $K \in \{4, \dots, 9\}$.

The contribution is a benchmark and metric centered on *compact faithfulness*. We introduce Complexity-Calibrated Local Concordance (CCC), which measures how well a top- K truncated additive explanation reconstructs the black-box probability. The project implements CCC together with standard comparison metrics and applies it to LIME, SHAP, and MAPLE on the UCI Breast Cancer Wisconsin Diagnostic and Adult datasets.

2 RELATED WORK

XAI evaluation is usually divided into human, application, and functional levels [4]. Human-grounded evaluation studies whether people can use explanations in simplified tasks; application-grounded

evaluation studies utility in a real domain workflow; and functionally-grounded evaluation uses computational proxies when human studies are not available. This project belongs to the third category: it evaluates formal properties of explanations produced for trained classifiers.

2.1 Local Explanation Methods

LIME fits a sparse local surrogate around the instance to be explained [10]. It is model-agnostic and intuitive, but its output can depend strongly on perturbation sampling, feature scaling, and the local linear fit. SHAP computes feature attributions based on Shapley values and satisfies desirable axioms for additive explanations [6]. In this project SHAP is implemented through KernelExplainer, using a sampled training background and the positive-class probability as the explained output. MAPLE learns local linear explanations by combining random-forest neighborhoods with supervised local regression [9]. In the codebase, MAPLE coefficients are converted into per-instance additive contributions by multiplying each coefficient by the corresponding feature value, so all three explainers share the same interface: weights and intercepts.

2.2 Evaluation Metrics

The literature review conducted for the project identified several families of functionally-grounded metrics. Fidelity metrics compare the explanation with the black-box model. Sufficiency and comprehensiveness perturb relevant features to check whether selected features preserve or remove prediction evidence. Robustness metrics, including sensitivity and infidelity [13], measure how explanations react to input perturbations. Benchmark projects such as XAI-Bench [5] and OpenXAI [1] systematize these criteria across methods and datasets. Frameworks such as LEAF [2] also include local concordance and conciseness criteria. However, these metrics usually treat faithfulness and compactness as separate properties: one metric asks whether the full explanation is faithful, another asks whether the explanation is small. CCC combines the two by asking whether the explanation is still faithful *after* enforcing a small top- K budget.

3 RESEARCH GAP

The project targets the gap between mathematical fidelity and usable fidelity. For an additive explanation $g(x) = w_0 + \sum_i \phi_i(x)$, a method may be perfectly faithful when every feature contribution is included. SHAP, for example, is designed to satisfy local accuracy under the full additive reconstruction. But a practitioner may show only the largest four to nine contributions in a dashboard, report, or interactive tool. If the omitted contributions collectively matter, the compact explanation can misrepresent the model even though the full explanation is exact.

This gap is not captured by model accuracy either. A high-accuracy classifier can be difficult to explain compactly, while a slightly less accurate model may produce local attributions that concentrate prediction evidence in fewer features. The research question is therefore:

Given a black-box model and a local additive explainer, how much fidelity is preserved when only the top- K feature contributions are shown?

Answering this question helps compare explainers under a realistic cognitive budget and provides a sanity check for feature rankings.

4 METHODOLOGY

4.1 Complexity-Calibrated Local Concordance

For each test instance x , the framework stores the black-box positive-class probability $f(x)$, the explainer intercept w_0 , and the additive contribution vector $\phi(x)$. Let $S_K(x)$ be the indices of the K largest absolute contributions. The truncated local reconstruction is

$$g_K(x) = w_0 + \sum_{i \in S_K(x)} \phi_i(x). \quad (1)$$

Complexity-Calibrated Local Concordance is the mean squared error between the black-box output and this truncated reconstruction:

$$CCC_K = \frac{1}{N} \sum_{j=1}^N \left(f(x^{(j)}) - g_K(x^{(j)}) \right)^2. \quad (2)$$

Lower CCC means that the explainer preserves local predictive fidelity using only K features.

The implementation intentionally does not multiply contributions by the raw features during reconstruction: SHAP values are already additive contributions, LIME returns local feature weights for the explained instance, and MAPLE coefficients are converted into contributions inside the explainer wrapper. This normalization is essential because CCC assumes a common additive form across methods.

4.2 Comparison Metrics

Table 1 summarizes the implemented metrics. Full MSE measures the fidelity of the complete additive explanation. Random- K MSE keeps K random contributions from the same weight vector and is repeated 30 times; it tests whether the explainer’s top- K ordering is better than arbitrary selection. Sufficiency and comprehensiveness use a feature baseline, implemented as the mean training vector, to evaluate prediction drift under input masking. The current codebase also includes normalized CCC and top- K degradation ratio for cross-dataset comparison and truncation analysis.

5 SOFTWARE ARCHITECTURE

The project is organized as a reproducible Python benchmark. The entry point `core/main.py` reads `config.yml`, builds a matrix of dataset–seed runs, executes them in parallel with `joblib`, and writes Markdown results. The experiment orchestration is implemented in `core/test_framework.py`. For each dataset and seed, the orchestrator: loads and scales the data, trains every configured model, generates explanations for every model/explainer pair, caches the

Table 1: Metrics implemented in `core/metrics.py`.

Metric	Quantity measured	Direction
CCC	Top- K reconstruction error	lower
Full MSE	Full additive reconstruction error	lower
Random- K	Random top- K baseline	lower
Sufficiency	Drift when only top- K inputs remain	lower
Comprehensive.	Drift when top- K inputs are removed	higher
Normalized CCC	CCC divided by prediction variance	lower
Degradation	CCC divided by full MSE	lower

Table 2: Mean model accuracy across the three seeds.

Dataset	Model	Accuracy
Adult	XGBoost	0.874
Adult	Neural network	0.821
Breast Cancer	XGBoost	0.968
Breast Cancer	Neural network	0.977

weights and intercepts, and computes every metric for all configured values of K .

The data loaders support the local copies of the UCI Breast Cancer Wisconsin Diagnostic and Adult datasets. Breast Cancer uses the WDBC files and recovers descriptive feature names from meta-data; Adult combines the train and test files, removes missing values, one-hot encodes categorical attributes, and standardizes features. The model wrappers expose a uniform `train`, `predict`, and `predict_proba` interface for XGBoost [3] and a scikit-learn multi-layer perceptron [8]. The visualization module generates summary charts from result files and per-instance top- K reduction figures for presentation.

6 EXPERIMENTAL SETUP

The main benchmark uses two public datasets: UCI Breast Cancer Wisconsin Diagnostic [11] and UCI Adult [12]. The classifiers are XGBoost and a neural network with two hidden layers of 100 units and a maximum of 500 training iterations. The explainers are LIME, SHAP KernelExplainer, and MAPLE. SHAP uses a sampled training background of size 100. MAPLE uses random forest feature extraction, 100 estimators, a validation size of 100, a training subsample of 2000 when available, and regularization 0.001. The benchmark evaluates $K = 4, 5, 6, 7, 8, 9$ over seeds 42, 123, and 2026.

The results discussed below come from the 20260629 archive file under `results/archive/`, which contains aggregate tables grouped by dataset, model, explainer, and K . The run reports six dataset–seed executions and includes model accuracy plus CCC, full MSE, sufficiency, comprehensiveness, and random- K MSE.

7 RESULTS AND ANALYSIS

Model accuracy, shown in Table 2, is not sufficient to select an explainer. On Breast Cancer, the neural network is slightly more accurate than XGBoost, but SHAP explanations for XGBoost reach lower compact error at small and large K . On Adult, XGBoost

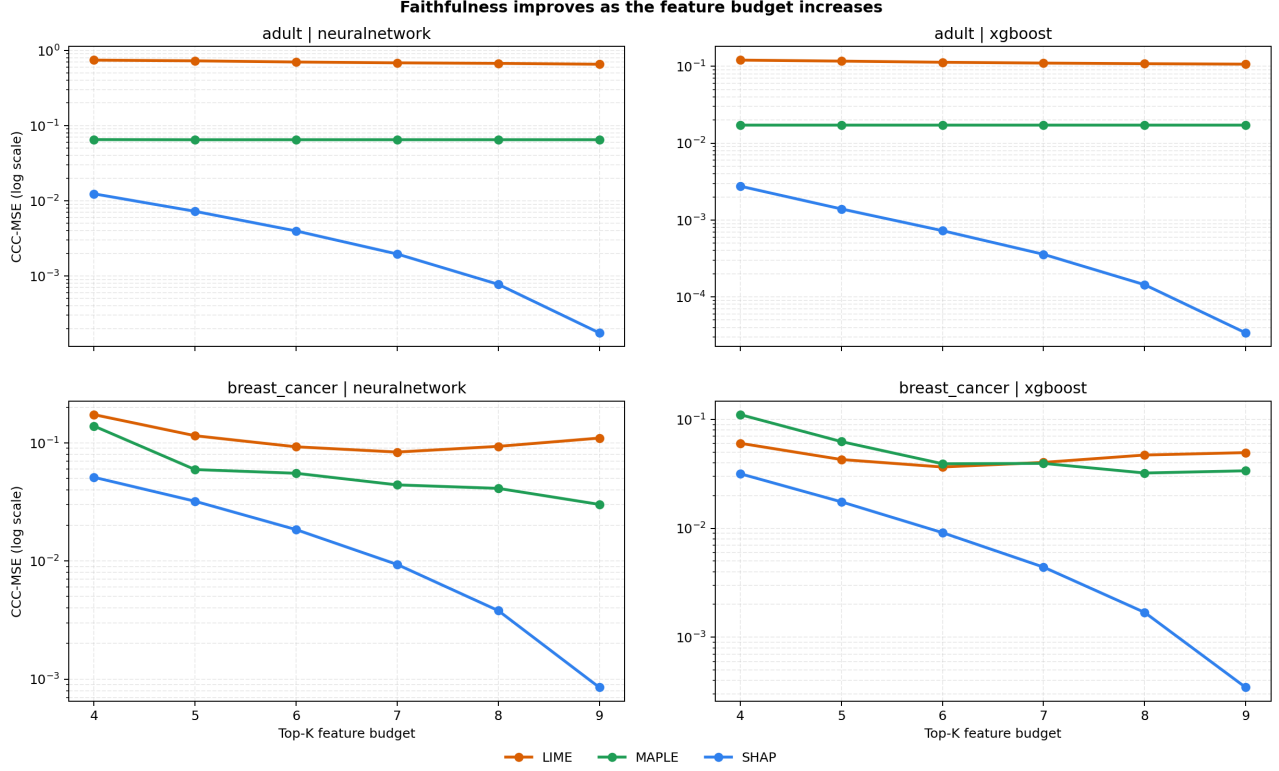


Figure 1: CCC as the feature budget increases. The figure was generated from the project visualization utilities. Lower curves indicate more faithful compact explanations.

Table 3: Representative aggregate CCC values from results_20260629_105230.md. Lower is better.

Dataset/model	Explainer	$K = 4$	$K = 5$	$K = 6$	$K = 7$	$K = 8$	$K = 9$
Adult/NN	SHAP	0.0125	0.0074	0.0040	0.0020	0.0008	0.0002
Adult/NN	MAPLE	0.0590	0.0590	0.0590	0.0590	0.0590	0.0590
Adult/NN	LIME	3.4195	3.3630	3.2186	3.0533	2.8499	2.6331
Adult/XGB	SHAP	0.0029	0.0014	0.0007	0.0003	0.0001	0.0000
Adult/XGB	MAPLE	0.0181	0.0180	0.0180	0.0179	0.0179	0.0179
Breast/XGB	SHAP	0.0286	0.0160	0.0082	0.0037	0.0014	0.0003
Breast/XGB	MAPLE	0.1210	0.0755	0.0556	0.0555	0.0405	0.0341
Breast/XGB	LIME	0.0518	0.0396	0.0401	0.0408	0.0468	0.0530

is more accurate than the neural network, but CCC still varies primarily with explainer choice rather than accuracy.

Figure 1 and Table 3 show three main patterns. First, SHAP is the strongest method under CCC. Its error decreases rapidly as K increases, reaching near-zero values at $K = 9$ on Adult/XGBoost and Breast Cancer/XGBoost. This is expected from SHAP’s additive local accuracy, but the non-zero values at small K are the important observation: a full SHAP explanation can be exact while its compact version still loses fidelity.

Second, MAPLE is an intermediate baseline, with behavior that depends on the dataset. On Breast Cancer/XGBoost, increasing K reduces CCC from 0.1210 to 0.0341, meaning that the top-ranked

contributions recover some predictive signal. On Adult/NN and Adult/XGBoost, however, MAPLE is almost flat across K . This suggests that the ranking does not concentrate the reconstruction signal in a small subset of features, even if the full local approximation is not terrible.

Third, LIME is unstable across settings. On Adult/NN, LIME CCC remains above 2.6 even at $K = 9$, several orders of magnitude worse than SHAP. On Breast Cancer/XGBoost, LIME is closer to SHAP for very small budgets but does not improve monotonically with K : the error decreases until about $K = 5$ or $K = 6$ and then rises again. This behavior indicates that adding more locally weighted features

does not necessarily make the truncated reconstruction closer to the black-box probability.

7.1 Top- K versus Random- K

The random- K baseline gives CCC an internal sanity check. In most tested settings, the explainer-selected top- K features outperform random subsets of the same explanation weights. For example, Adult/XGBoost SHAP at $K = 9$ has CCC around 3.1×10^{-5} , while random- K MSE is about 0.086. Breast Cancer/XGBoost SHAP at $K = 9$ has CCC around 0.00027, while random- K is about 0.113. This confirms that the top-ranked features preserve meaningful prediction signal. When the gap is smaller, as for some MAPLE configurations, the feature ranking is less informative.

7.2 What CCC Reveals

The main insight is that CCC separates three notions that can otherwise be confused: model accuracy, full local fidelity, and compact local fidelity. SHAP’s full MSE is approximately zero because the full additive reconstruction matches the explained model output, but CCC still exposes the cost of showing only four or five features. MAPLE can have moderate full MSE but poor top- K concentration. LIME can produce local weights whose selected features do not reconstruct the probability well. These distinctions are directly relevant for systems that display explanations to humans, because those systems normally display only a subset of features.

8 LIMITATIONS

CCC applies naturally to additive explanations, but it is not directly suitable for rule lists, counterfactual sets, prototypes, or explanations whose output is not a decomposable sum. The metric also depends on the scale and calibration of the model probability. For this reason the code includes normalized CCC, which divides the error by the variance of black-box probabilities, although the main archived run discussed here reports raw CCC. Perturbation metrics such as sufficiency and comprehensiveness depend on the masking baseline; the current implementation uses the mean training vector, which is simple and reproducible but may create unrealistic inputs for correlated features. Finally, the benchmark is limited to two datasets and two model families. The conclusions are therefore strongest as evidence for the proposed evaluation protocol, not as a universal ranking of all XAI methods.

9 REPRODUCIBILITY AND SUBMISSION MATERIAL

The implementation is contained in the submitted project archive and in the Git repository <https://github.com/itsPinguiz/EQE>. The main entry point is `core/main.py`; configuration is in `config.yml`; the archived 20260629 result file is under `results/archive/`; and the extended technical notes are in `docs/relazione.md`. The public datasets are referenced above and are also included locally under `datasets/`. Slides are optional for submission; generated figures under `results/figures/` can be reused for the final presentation.

10 FUTURE WORK

The most immediate extension is to broaden the benchmark without changing the metric definition. Additional tabular datasets would test whether CCC remains stable when the number of features, categorical sparsity, and class imbalance change. Additional black-box models, such as calibrated linear models, random forests, and gradient-boosted variants, would also clarify whether compact faithfulness depends more on the explainer or on the geometry of the learned decision function.

A second direction is to make the feature budget adaptive. The current report uses fixed values from four to nine because they are easy to compare and align with a cognitive-complexity motivation. In an interface, however, a useful system could choose the smallest K whose CCC falls below a model-specific threshold, or expose a warning when no compact explanation reaches acceptable fidelity. This would turn CCC from an offline benchmark metric into a runtime quality-control signal for deployed explanation tools.

11 CONCLUSIONS

EQE contributes a compact-faithfulness evaluation of local explanations through Complexity-Calibrated Local Concordance. CCC complements existing XAI metrics by asking whether an additive explanation remains faithful after imposing a human-sized feature budget. The benchmark shows that SHAP is the strongest method under this criterion, MAPLE is a useful intermediate comparison, and LIME is less reliable in the tested configurations. More importantly, the results show that full-explanation fidelity and compact-explanation fidelity are distinct evaluation targets. This makes CCC useful for practical XAI settings where explanations must be both faithful to the model and short enough to be inspected by a human.

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